Ridwan Olalekan Olabiyi

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Research Interests

Scientific machine learning; Bayesian inference; stochastic inverse problems; uncertainty quantification; governing-physics discovery; causal discovery; neural/operator learning; image-based 3D reconstruction; photogrammetry; photometric stereo; statistical quality engineering; advanced manufacturing; materials characterization; semiconductor systems.

Education

Ph.D., Industrial Engineering, Arizona State University Expected Spring 2027
Advisor: Ashif Iquebal.
Working dissertation title: *Physics-Based Machine Learning for Inverse Problems and Scientific Discovery under Uncertainty.*

M.S., Industrial Engineering, Arizona State University December 2024
Advisor: Ashif Iquebal. Degree GPA: 3.73.
Thesis: *A Bayesian Spatiotemporal Modeling Approach to the Inverse Heat Conduction Problem.*

B.Sc., Agricultural & Environmental Engineering, Obafemi Awolowo University 2019
First-Class Honors; CGPA 4.58/5.00; class rank 2/25.

Venture in Management Programme, Lagos Business School, Pan-Atlantic University 2020

Academic Appointments & Research Experience

Research Associate, School of Computing and Augmented Intelligence, Arizona State University Current
Develop physics-aware and uncertainty-aware computational methods for inverse problems, scientific discovery, and computationally efficient engineering modeling. Current work spans advanced manufacturing, materials characterization, dynamical systems, scientific machine learning, image-based quality assurance, and collaborations with researchers at NIST and Oak Ridge National Laboratory.

Graduate Research Associate, Arizona State University 2021–2022
Research under Ashif Iquebal in Bayesian inference, inverse problems, data-driven engineering, and scientific machine learning.

Graduate Teaching Assistant, Arizona State University 2022–2023
Supported IEE 380, *Probability and Statistics for Engineering Problem Solving*, in Fall 2022 and Spring 2023.

Selected Research Contributions

- **Physics discovery under uncertainty:** Developed a stochastic inverse framework for recovering governing equations in the presence of measurement noise and intrinsic parameter variability; published in *Communications Physics* (2026).
- **Bayesian spatiotemporal inverse heat conduction:** Developed an uncertainty-aware framework for reconstructing latent thermal quantities from indirect temperature measurements; conference publication (2023) and journal publication (2024).
- **Stochastic inverse indentation:** Developed inverse methodologies for elastoplastic property recovery from indentation data; current extension addresses manufacturing-induced variability and measurement uncertainty with NIST/ORNL collaborators.
- **CRONet and topology optimization:** Developed neural/operator-learning approaches for accelerating iterative topology optimization; published in *Manufacturing Letters* and extended through FPGA/AIE-ML acceleration research at IEEE FCCM 2026.
- **Image-based 3D reconstruction for AM quality assurance:** Developing a non-contact inspection system for additively manufactured components that combines controlled image acquisition, multi-view photogrammetry, photometric stereo, calibration, surface reconstruction, point-cloud fusion, and CAD/reference comparison. Current work focuses on robust reconstruction and positive surface-deviation detection; longer-term integration targets post-print inspection, in-situ quality monitoring, and targeted surface processing on a Meltio robotic metal-AM platform.
- **Causal discovery:** Ongoing work on nonlinear dynamical-system discovery through counterfactual explanations and causal reasoning.

Journal & Archival Publications

- Christy Dunlap, **Ridwan Olabiyi**, Firas Al-Hindawi, Hari Pandey, Stephen Pierson, Daniel Curl, Braden Stevens, Mohammad Ishraq Hossain, Annapurna Parjuli, Chinmaya Joshi, Ashif Iquebal, and Han Hu. "Open datasets and machine learning for two-phase heat transfer: a review following a spatial-temporal taxonomy." *Transport Phenomena*, 2026. doi:10.1515/tp-2026-0081.
- **Ridwan Olabiyi**, Han Hu, and Ashif Iquebal. "Discovering Governing Physics in the Presence of Uncertainty." *Communications Physics*, 2026. doi:10.1038/s42005-026-02711-7.
- **Ridwan Olabiyi**, Hui Yang, and Ashif Iquebal. "CRONet: A Convolutional Recurrent Operator Approximator Network to Accelerate Topology Optimization." *Manufacturing Letters*, 2025. doi:10.1016/j.mfglet.2025.06.125.
- **Ridwan Olabiyi**, Hari Pandey, Han Hu, and Ashif Iquebal. "A Bayesian Spatiotemporal Modeling Approach to the Inverse Heat Conduction Problem." *ASME Journal of Heat and Mass Transfer*, 146(9), 091403, 2024. doi:10.1115/1.4065451.

Peer-Reviewed Conference Publications

- Kaustubh Mhatre, Vedant Tewari, Aditya Ray, Farhan Khan, **Ridwan Olabiyi**, Ashif Iquebal, and Aman Arora. "Accelerating Topology Optimization on AMD Versal AIE-ML Engines." *2026 IEEE 34th Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM)*, 2026. doi:10.1109/FCCM68464.2026.00047.
- **Ridwan Olabiyi**, Jordan Weaver, and Ashif Iquebal. "Characterization of Elastoplastic Properties of Additively Manufactured Specimens from Indentation Data Using Stochastic Inverse Modeling." *ASME 2024 International Manufacturing Science and Engineering Conference*, 2024. doi:10.1115/MSEC2024-125452.
- **Ridwan Olabiyi**, Hari Pandey, Han Hu, and Ashif Iquebal. "A Bayesian Spatio-Temporal Modeling Approach to the Inverse Heat Conduction Problem." *ASME 2023 Heat Transfer Summer Conference*, 2023. doi:10.1115/HT2023-107671.

Preprints & Open Review

- **Ridwan Olabiyi**, Ishraq R. Rahman, and Ashif Iquebal. "CaNDiCE: Causal Discovery of Nonlinear Dynamics Through Counterfactual Explanations." OpenReview, ICLR 2026 submission. OpenReview.
- Kaustubh Mhatre, Vedant Tewari, Aditya Ray, Farhan Khan, **Ridwan Olabiyi**, Ashif Iquebal, and Aman Arora. "Accelerating CRONet on AMD Versal AIE-ML Engines." arXiv:2604.14700, 2026. arXiv.
- **Ridwan Olabiyi**, Han Hu, and Ashif Iquebal. "Discovering Governing Equations in the Presence of Uncertainty." arXiv:2507.09740, 2025. arXiv.

Manuscript in Preparation

- **Ridwan Olabiyi**, Jordan S. Weaver, Patxi Fernandez-Zelaia, and Ashif Iquebal. "Accelerating Elastoplastic Property Characterization Under Manufacturing-Induced Variability Through Stochastic Inverse Modeling of Indentation Data." Manuscript in preparation for an IISE journal submission, 2026.

Selected Conference Presentations

- INFORMS Annual Meeting Job Market Showcase, San Francisco, 2026: *Stochastic Inverse Modeling for Rapid Elastoplastic Property Characterization of Additively Manufactured Components.*
- NAMRC 53, Greenville, SC, 2025: *CRONet: A Convolutional Recurrent Operator Approximator Network to Accelerate Topology Optimization.*
- IISE Annual Conference, Atlanta, 2025: stochastic inverse modeling for elastoplastic characterization; technical and poster presentations.
- INFORMS Annual Meeting, Seattle, 2024: *Consistent Discovery of Dynamical Systems Using Stochastic Inverse Modeling*; QSR Best Student Paper finalist presentation.
- ASME MSEC, Knoxville, 2024: stochastic inverse indentation for additively manufactured specimens.
- INFORMS Annual Meeting, Phoenix, 2023: Bayesian spatiotemporal inverse heat-conduction research; technical and poster presentations.
- ASME Heat Transfer Summer Conference, Washington, DC, 2023: Bayesian spatiotemporal inverse heat conduction.
- INFORMS Annual Meeting, Indianapolis, 2022: *Minimal Subset Learning in Federated Smart Manufacturing.*

Teaching Experience

IEE 570 – Advanced Quality Control, Arizona State University Summer 2026
Graduate-level instruction, student consultations, assessment, exam review, and course grading.

IEE 470 – Quality Control, Arizona State University Spring 2026

Undergraduate quality-control instruction connecting statistical methods with engineering process-improvement decisions.

IEE 570 – Advanced Quality Control, Arizona State University Summer 2024

Instructional support in advanced statistical quality-control concepts and applications.

IEE 380 – Probability and Statistics for Engineering Problem Solving, ASU Fall 2022, Spring 2023

Graduate Teaching Assistant.

Volunteer GRE Instructor, Nigeria Oct. 2020–July 2021

Quantitative reasoning and standardized-test preparation for prospective graduate students.

Classroom Teacher, City College Enugu, Nigeria 2020–2021

Mathematics, Physics, Basic Science, and Technology.

Course Tutor, Brainfill Academy, Nigeria 2020

Advanced-level mathematics including algebra, trigonometry, geometry, and calculus.

Research Mentoring

- Mentor a recent Mechanical Engineering graduate and an Industrial Engineering Ph.D. student on an image-based 3D reconstruction system for additive-manufacturing quality assurance. The project combines controlled imaging, photogrammetry, photometric stereo, calibration, surface reconstruction, and reference-geometry comparison, with longer-term integration targeting robotic metal-AM inspection and targeted post-processing; the collaboration has also included award-winning engineering competitions.
- Mentored an international undergraduate researcher from France through ASU's Summer Research Initiative on active-learning research.
- Co-mentored two master's students alongside Ashif Iquebal on generative CAD generation from hand-drawn sketches and inverse modeling for microstructural characterization using eddy-current field data.
- Provide recurring research discussion, technical feedback, troubleshooting, and methodological guidance to additional master's and Ph.D. researchers.

Honors, Awards & Fellowships

- First Place, NSF Future Manufacturing Data Challenge, 2026.
- Best Presentation Design Award, NSF Future Manufacturing Data Challenge, 2026.
- Winner, Semiconductor Solutions Challenge, Problem D, 2026.
- Trailblazers in Engineering Fellow, Purdue University College of Engineering, 2025.
- National Science Foundation Travel Award, NAMRC 53 / MSEC 2025.
- Harold & Lucille Dunn Memorial Engineering Scholarship, 2025, 2024, 2023, 2022.
- Graduate College Travel Award, IISE Annual Conference, 2025.
- SCAI IE Doctoral Fellowship, 2025 and 2024.
- Finalist, INFORMS Quality, Statistics & Reliability Best Student Paper Competition, 2024.
- Graduate Student Government Travel Grant, INFORMS Annual Meeting, 2024.
- SCAI Conference Funding, INFORMS Annual Meeting, 2024 and 2023.
- NSF Early Career Researcher Travel Award, MSEC 2024.
- University Graduate Fellowship, Arizona State University, 2022.
- Amazon SCOT / INFORMS Scholarship, 2022.
- CIDSE Doctoral Fellowship, Arizona State University, 2021.
- Male Best Speaker, NYSC Enugu Orientation Camp, 2020.
- Venture in Management Programme Scholarship, Lagos Business School, 2020.
- Overall Best Graduating Student (Male), Obafemi Awolowo University, 2019.
- Best Graduating Student, Agricultural & Environmental Engineering, OAU, 2019.
- Professor V. A. Oyenuga Prize, OAU, 2019.
- Engineer E. A. Ladipo's Prize, OAU, 2019.
- Abdulkabir Aliu Foundation Undergraduate Scholarship, 2016–2018.

Conference Leadership & Professional Service

- Co-Organizer, invited Data Mining cluster session, *Data-Driven Modeling and Scientific Discovery in Complex Systems*, INFORMS Annual Meeting, 2026.

- Session Chair, INFORMS Annual Meeting, 2025.
- Session Chair, Additive Manufacturing technical session, NAMRC 53 / MSEC, 2025.
- Session Chair, *Data-driven Models for Scientific Discovery*, INFORMS Annual Meeting, 2024.

Journal Peer Review: *Computers & Industrial Engineering*; *Engineering Applications of Artificial Intelligence*; *Journal of Manufacturing Processes*; *The Journal of Supercomputing*.
Conference Peer Review: ASME MSEC (2024–2025); NAMRC (2024–2025); IISE Annual Conference & Expo (2025).

Leadership, Outreach & Volunteering

- Vice President, ASU INFORMS Student Chapter, 2023–2025.
- Resource Person, EducationUSA Graduate Boot Camp, 2021 and 2024.
- Student Volunteer, INFORMS Annual Meeting, Phoenix, 2023.
- Google Road Show Internet-Safety Outreach, Lagos State, 2020.
- Volunteer, Junior Achievement Nigeria, 2019–2020.
- Agro-Allied Community Development Service, 2019–2020.
- Chairman, Best Brain Challenge Committee, Obafemi Awolowo University, 2019.
- Prospective Corps Members Welcoming & Guidance Team, Muslim Corpers Association of Nigeria, 2019.
- Co-convener, Faculty of Technology Muslim Students Association Annual Symposium, OAU, 2018.
- Departmental Football Coach, OAU, 2017–2018.
- Executive Member, Student Representative Council, AGEESS, OAU, 2015–2018.
- Course Tutor, Muslim Student Society of Nigeria, OAU Chapter, 2014–2016.

Professional Memberships

INFORMS; Institute of Industrial and Systems Engineers (IISE); American Society of Mechanical Engineers (ASME); Society of Manufacturing Engineers (SME); Society for Industrial and Applied Mathematics (SIAM); Nigerian Society of Engineers (NSE); Nigerian Institution of Agricultural Engineers (NIAE).

Technical Skills & Certifications

Programming/Software: Python, MATLAB, Jupyter, Git/GitHub, AMPL, IBM CPLEX, ABAQUS, COMSOL Multiphysics.
Methods: Bayesian inference, stochastic inverse modeling, uncertainty quantification, design of experiments, causal inference, scientific machine learning, neural/operator learning, statistical quality control, finite-element modeling, image-based 3D reconstruction, photogrammetry, photometric stereo, point-cloud processing, and CAD-referenced surface inspection.
CITI Program: Human Research: IRB – Biomedical Research (Group 1), completed 2026, valid through August 2030; Responsible Conduct of Research, Graduate Student and Postdoctoral Researcher RCR, completed 2024.

Prior Engineering Experience

Assistant Program Engineer (NYSC), Enugu State Agricultural Development Program 2019–2020
Supported agricultural-equipment management, technical training for farmers, and engineering-program operations.
Workshop / Production Engineering Intern, International Institute of Tropical Agriculture 2017–2018
Supported maintenance and operation of agricultural machinery, workshop equipment, and production/packaging systems.

Selected Media & Professional Profiles

- Purdue University College of Engineering – 2025 Trailblazers in Engineering.
- The Guardian – NSF-supported manufacturing research.
- The Guardian – INFORMS/QSR recognition.
- Amazon Science – 2022 Amazon SCOT / INFORMS Scholars.